What are the features of python.

Q) What are the applications of python

⇒ * web development → building websites

* data science & data Analysis → handling and analyzing data.

* Artificial Intelligence & Machine learning → smart systems and predictions

* Automation and scripting → performing repetitive tasks

* Game development → creating simple games

* desktop Application development → software development

* Cyber security & ethical Hacking →

* scientific and Mathematical computing → research and calculation

* Internet of things (IOT) → TO control smart devices & sensors.

* cloud computing → used in cloud platforms and services.

Q) What are the features of python.

Features of python :

* Easy to learn → simple and Readable syntax

* Interpreted language → no need to compile.

* platform Independent → runs on all operating systems.

* dynamically typed → no need to declare data types.

* Object oriented → supports OOP concepts

* Large library support → many built-in modules

* open sources → free to use and modify

* High-level language → closer to human language.

Q) What is python programming ?

Python is a high level, easy-to-learn programming language used to build software, websites, apps and more.

* Easy to read and write (simple syntax)
* works on different platforms (Windows, Mac, Linux)
* used by beginners and professionals.
* supports multiple programming styles (like object-oriented and functional)
* large community support - Many developers use python, so help and resources are easily available.
* Rich libraries and frameworks - python has many built-in libraries that make coding faster and easier.

Q) What are the different types of errors python.

There are three types of errors

* Syntax errors
* Runtime Errors
* Logical errors.

1) Syntax errors :-

Mistakes in writing code (wrong grammar of python)

* Missing colon, brackets, or wrong indentation.

2) Runtime errors :-

Errors that occur while the program is running.

* division by zero, file not found.

3) Logical errors :-

program runs, but gives wrong output.

* wrong formula or condition used.

Write the how many modes we can execute python program.

there are two modes are in python program.

* Interactive mode
* Script mode.

Interactive mode: Typically used to learn or test python.

* Execute code line by line.

* Write code directly in python shell and get instant outp

Script Mode:

* Execute entire program at once
* Write code in a file (.py) and run it.
* Typically used for robotics or other projects.

Q) Explain about print function.

* The print () function is used to display output on the screen.
* used to show text or results.
* outputs data to the console (screen)
* can print multiple values.
* Supports formatting (using sep and end)
* sep → separates values
* end → what to print at the end
* syntax:

print (object(s), sep=' ', end='\n')

Q) Explain about assignment statements.

* An assignment statement is used to assign (store) a value to a variable.
* uses = (assignment operator)
* can assign multiple values at once.
* supports updating values
* value can be a number, text, or result of expression.

* variables are created automatically during assignment.
* No need to declare variable type.
* values can be changed anytime.
* supports swapping values.

Q) Explain about selective statements.

A selective statement is used to make decisions in a program based on conditions.

* Helps control program flow
* uses keywords like if, else, elif
* executes code based on condition.

* Types of selective statements.

→ If statement → executes when condition is true

→ If-else statement → chooses between two options

→ If-else-elif ladder → Multiple conditions checking

→ Nested if → if inside another if.